

HR Management System - Functional Specification

This document outlines the functional requirements of the **HR Management System (HRMS)**, including **CRUD operations**, **validation rules**, **filtering/search capabilities**, and **business logic**.

1. Functional Modules

The system consists of the following core modules:

1. **Location Management**
2. **Department Management**
3. **Job Role Management**
4. **Employee Management**
5. **Job History Tracking**
6. **Advanced Search & Filtering**

Each module allows standard **CRUD (Create, Read, Update and Delete)** operations with specific validation rules.

2. Functional Details

2.1 Location Management

CRUD Operations:

- Create, edit, delete, and list locations.

Validation Rules:

- Name: **Required**, unique per location.
 - City: **Required**.
 - Country: **Required**.
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2.2 Department Management

CRUD Operations:

- Create, edit, delete, and list departments.

Validation Rules:

- Name: **Required**, unique within the same location.
 - Description: **Required**.
 - LocationId: **Required**, must reference an existing location.
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2.3 Job Role Management

CRUD Operations:

- Create, edit, delete, and list job roles.

Validation Rules:

- Title: **Required**, unique within the same department.
 - Description: **Required**.
 - DepartmentId: **Required**, must reference an existing department.
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2.4 Employee Management

CRUD Operations:

- Create, edit, delete, and list employees.

Validation Rules:

- Name: **Required**.
- Gender: **Required** (Male / Female / Other).
- BirthDate: **Required**, must not be in the future.
- Phone: **Required, unique** (no duplicate phone numbers).
- Email: **Required, unique** (no duplicate emails).
- Address: **Required**.
- Status: **Required** (Active, Inactive, Resigned, On Leave).

Business Logic:

- Employees **cannot be deleted** if they have active job history records.
 - Only **Admin Users** can delete employee records.
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2.5 Job History Tracking

CRUD Operations:

- Create, edit, delete, and list job history records.

Validation Rules:

- EmployeeId: **Required**, must reference an existing employee.
- JobRoleId: **Required**, must reference an existing job role.
- StartDate: **Required**.
- EndDate: **Optional**, must be **greater than** StartDate (if provided).
- Status: **Required** (Active, Completed, Terminated).
- ManagerId: **Optional**, must reference an existing employee.

Business Logic:

- An **employee can only have one active job at a time**.
 - If a **job history is deleted**, it should not affect past records.
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3. Search & Filtering Operations

The system supports **advanced search and filtering** based on the following criteria:

3.1 Employee Search & Filtering

Users can search employees by:

- **Employee Name** (LIKE search for partial matches)
- **Department**
- **Gender**
- **Status** (Active, Inactive, Resigned, On Leave)
- **Start Date & End Date**

Start Date & End Date Filtering Logic:

Scenario	Expected Result
Start Date only provided	Returns employees whose job history started on or after the given date.
End Date only provided	Returns employees whose job history ended on or before the given date.
Both Start & End Date provided	Returns employees whose job history falls within the date range .
Neither Start nor End Date provided	Returns all employees .